Fire-Aware Robot Dogs

Machine-Learning Wildfire Forecasts for Safe Navigation and Scouting

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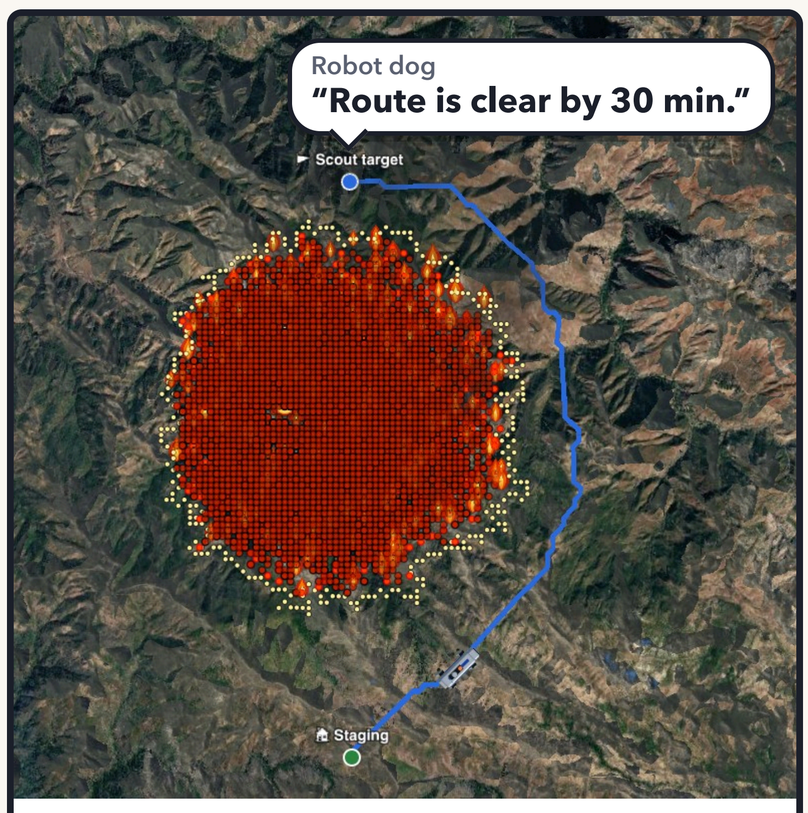
One afternoon at a Big Sur fire

A simulated mission. Every picture is from our simulator.



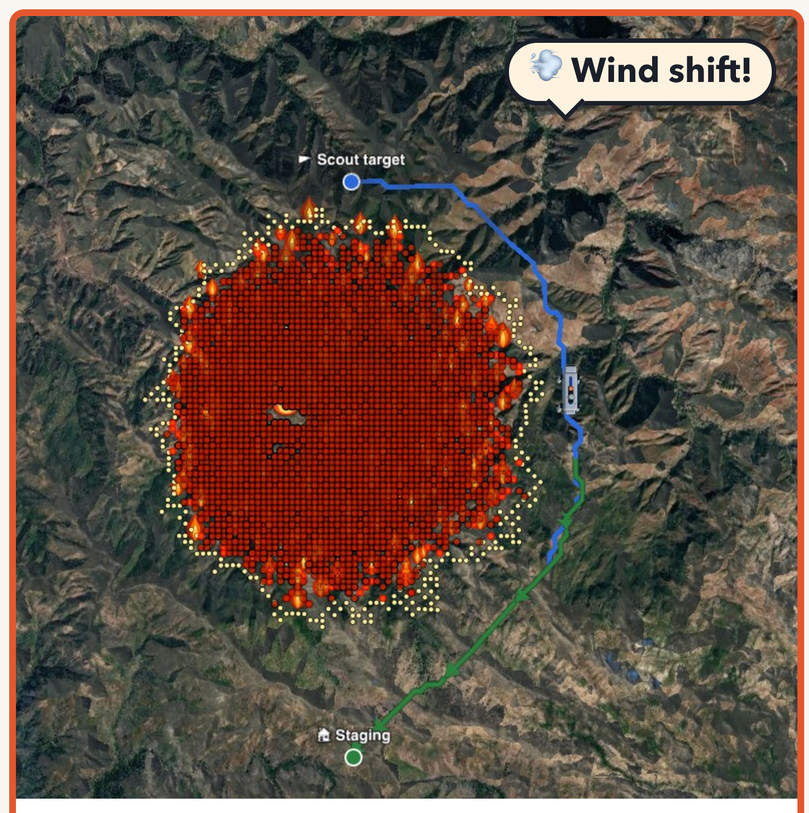
3:00 PM

The fire has burned since midnight on steep, brushy slopes. The crew needs to know **where the edge is going**.



3:00 PM • Plan

On its forecast, the dog plans a route (**blue**) that passes every spot **30+ min before the fire**, slower through brush and uphill.



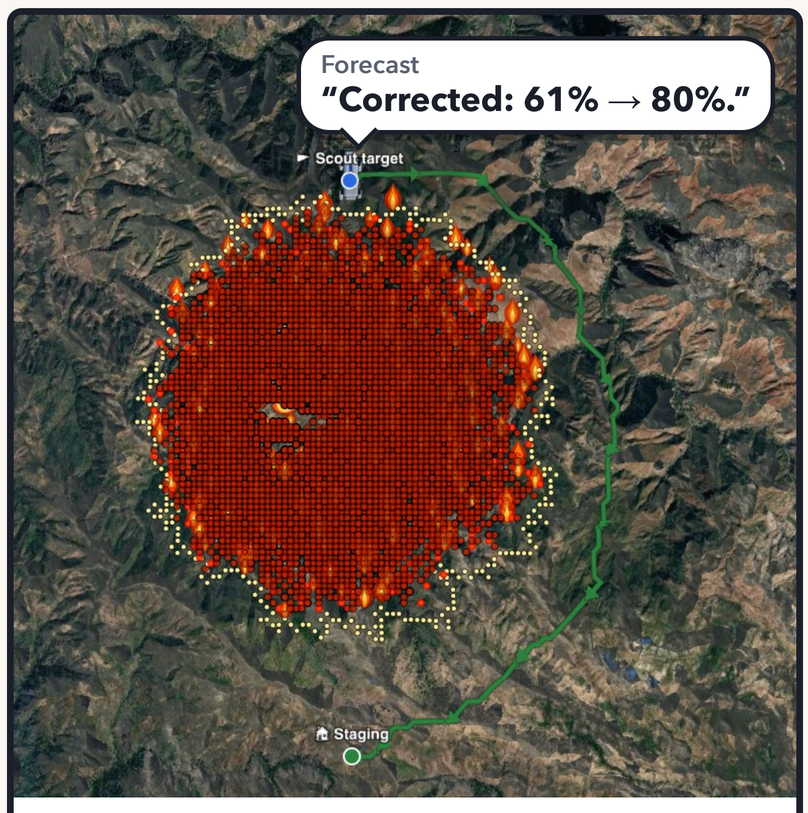
3:30 PM • Surprise

The wind shifts and the real fire turns. **The forecast doesn't know**: match 82% → 61%. The escape path home (**green**) stays open.



3:45 PM • Measure

Its sensors see the real fire coming. It measures **wind, temperature, humidity and the fire edge** and sends them back.



4:45 PM • Scout

On the **corrected forecast** the dog re-plans, reaches the fire edge and scouts it.

✓ Home safe

0 people sent to the fire edge

Forecast match **61% → 80%**

Fire margin kept: **≥ 30 min**

~8 PM • Home

Back at staging before the fire arrives. The crew plans with a **better forecast**.

The loop behind the story: **Predict** → **Safe route** → **Escape path** → **Scout** → **Feedback** ↺

Know where the fire *will* be – not just where it is.

Safer crews
Fewer firefighters sent into danger to look.

Robot stays safe
Routes and exits keep it from getting trapped.

Smarter sensing
Measure where it changes decisions.